ABCD ADHD menarche analysis: Output for longitudinal models

Model 2 longitudinal no interaction output

```
   Estimate Est.Error  Q2.5 Q97.5
R2    0.827   0.00459 0.818 0.836


 Family: poisson
  Links: mu = log
Formula: internalising_wave_3 ~ menarche_status_p_wave_2 + adhd_status_life + age_wave_2_sc +
   Data: analytic_df_wide (Number of observations: 4496)
  Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~family_id (Number of levels: 3915)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.840     0.015    0.811    0.870 1.000     1150     2010

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.109     0.032    0.051    0.179 1.003      518      589

Regression Coefficients:
                          Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS
Intercept                    1.191     0.037    1.116    1.260 1.000     1359
menarche_status_p_wave_2Y    0.175     0.033    0.109    0.238 1.001     1113
adhd_status_lifeY            0.242     0.033    0.176    0.305 1.001     1158
age_wave_2_sc               -0.008     0.015   -0.037    0.023 1.002     1194
ethnraceblack_nh            -0.402     0.056   -0.509   -0.293 1.000     1074
ethnracehispanic            -0.094     0.050   -0.193    0.004 1.000      745
ethnraceother                0.001     0.051   -0.100    0.099 1.002      796
```

```
inr_wave_0_sc               0.042     0.017    0.009    0.076 1.003     1127
internalising_wave_0_sc     0.417     0.014    0.391    0.444 1.002      778
                         Tail_ESS
Intercept                    1966
menarche_status_p_wave_2Y    1927
adhd_status_lifeY            1650
age_wave_2_sc                1460
ethnraceblack_nh             1899
ethnracehispanic             1487
ethnraceother                1646
inr_wave_0_sc                2027
internalising_wave_0_sc      1415

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

Model 2 longitudinal no interaction effects expressed as % change in expected outcome per 1-unit increase in original predictor scale

```
# A tibble: 2 x 4
  Variable            Estimate `Lower 95% CI` `Upper 95% CI`
  <chr>                  <dbl>          <dbl>          <dbl>
1 Menarche Y (Wave 2)     19.2           11.6           26.9
2 ADHD Y                  27.4           19.3           35.6
```

Model 2B longitudinal no interaction output

```
   Estimate Est.Error  Q2.5 Q97.5
R2    0.827   0.00453 0.819 0.836


 Family: poisson
  Links: mu = log
Formula: internalising_wave_3 ~ breast_development_p_wave_2_sc + adhd_status_life + age_wave_
   Data: analytic_df_wide (Number of observations: 4515)
  Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~family_id (Number of levels: 3928)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
```

```
sd(Intercept)    0.846     0.015    0.818    0.875 1.001     1111     2368

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.111     0.032    0.055    0.182 1.002      628      908

Regression Coefficients:
                               Estimate Est.Error l-95% CI u-95% CI  Rhat
Intercept                         1.264     0.036    1.191    1.331 1.001
breast_development_p_wave_2_sc    0.077     0.015    0.048    0.108 1.001
adhd_status_lifeY                 0.226     0.033    0.162    0.292 1.003
age_wave_2_sc                     0.007     0.015   -0.023    0.036 1.001
ethnraceblack_nh                 -0.392     0.057   -0.501   -0.280 1.001
ethnracehispanic                 -0.074     0.048   -0.170    0.020 1.000
ethnraceother                    -0.009     0.050   -0.106    0.087 1.000
inr_wave_0_sc                     0.032     0.017   -0.001    0.066 1.001
internalising_wave_0_sc           0.418     0.014    0.391    0.445 1.001
                               Bulk_ESS Tail_ESS
Intercept                          1203     1400
breast_development_p_wave_2_sc     1270     1660
adhd_status_lifeY                  1381     2246
age_wave_2_sc                      1474     1980
ethnraceblack_nh                   1105     1899
ethnracehispanic                   1155     1894
ethnraceother                      1156     1902
inr_wave_0_sc                      1223     1732
internalising_wave_0_sc             970     1884

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

Model 2B longitudinal no interaction effects expressed as % change in expected outcome per 1-unit increase in original predictor scale

```
# A tibble: 2 x 4
  Variable                    Estimate `Lower 95% CI` `Upper 95% CI`
  <chr>                          <dbl>          <dbl>          <dbl>
1 Breast development (Wave 2)     6.08           3.75            8.7
2 ADHD Y                         25.3           17.6            34.0
```

Model 3 longitudinal no interaction output

```
   Estimate Est.Error  Q2.5 Q97.5
R2    0.823   0.00528 0.812 0.833


 Family: poisson 
  Links: mu = log 
Formula: externalising_wave_3 ~ menarche_status_p_wave_2 + adhd_status_life + age_wave_2_sc +
   Data: analytic_df_wide (Number of observations: 4496) 
  Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
         total post-warmup draws = 8000

Multilevel Hyperparameters:
~family_id (Number of levels: 3915) 
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    1.012     0.020    0.972    1.052 1.001     2383     4460

~site_id (Number of levels: 22) 
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.061     0.036    0.004    0.139 1.001      750     2098

Regression Coefficients:
                          Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS
Intercept                    0.342     0.038    0.267    0.416 1.000     4201
menarche_status_p_wave_2Y    0.131     0.041    0.052    0.211 1.000     4086
adhd_status_lifeY            0.579     0.043    0.490    0.661 1.000     3580
age_wave_2_sc               -0.051     0.020   -0.090   -0.013 1.001     4133
ethnraceblack_nh            -0.099     0.067   -0.231    0.030 1.001     3273
ethnracehispanic            -0.089     0.058   -0.203    0.025 1.003     3676
ethnraceother               -0.035     0.065   -0.163    0.091 1.000     3640
inr_wave_0_sc                0.001     0.022   -0.041    0.043 1.000     3797
externalising_wave_0_sc      0.483     0.017    0.449    0.517 1.001     2428
                          Tail_ESS
Intercept                     5616
menarche_status_p_wave_2Y     5285
adhd_status_lifeY             4704
age_wave_2_sc                 4687
ethnraceblack_nh              4298
ethnracehispanic              4976
ethnraceother                 4844
inr_wave_0_sc                 4977
externalising_wave_0_sc       4318
```

```
Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

Model 3 longitudinal no interaction effects expressed as % change in expected outcome per 1-unit increase in original predictor scale

```
# A tibble: 2 x 4
  Variable            Estimate `Lower 95% CI` `Upper 95% CI`
  <chr>                  <dbl>          <dbl>          <dbl>
1 Menarche Y (Wave 2)     14.0            5.3           23.5
2 ADHD Y                  78.4           63.3           93.6
```

Model 3B longitudinal no interaction output

```
   Estimate Est.Error  Q2.5 Q97.5
R2    0.825   0.00514 0.815 0.835


 Family: poisson
  Links: mu = log
Formula: externalising_wave_3 ~ breast_development_p_wave_2_sc + adhd_status_life + age_wave_
   Data: analytic_df_wide (Number of observations: 4515)
  Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
         total post-warmup draws = 8000

Multilevel Hyperparameters:
~family_id (Number of levels: 3928)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    1.018     0.020    0.979    1.057 1.001     2580     4284

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.057     0.035    0.004    0.133 1.003      683     2247

Regression Coefficients:
                               Estimate Est.Error l-95% CI u-95% CI  Rhat
Intercept                         0.399     0.035    0.329    0.469 1.001
breast_development_p_wave_2_sc    0.090     0.019    0.053    0.128 1.001
adhd_status_lifeY                 0.564     0.044    0.479    0.650 1.000
age_wave_2_sc                    -0.054     0.019   -0.093   -0.016 1.002
```

```
ethnraceblack_nh                 -0.107     0.067   -0.241    0.022 1.000
ethnracehispanic                 -0.081     0.058   -0.194    0.033 1.001
ethnraceother                    -0.051     0.065   -0.178    0.078 1.000
inr_wave_0_sc                    -0.006     0.022   -0.048    0.037 1.001
externalising_wave_0_sc           0.483     0.017    0.449    0.517 1.001
                               Bulk_ESS Tail_ESS
Intercept                          3906     4763
breast_development_p_wave_2_sc     4273     5409
adhd_status_lifeY                  3185     4805
age_wave_2_sc                      3977     4877
ethnraceblack_nh                   2870     4384
ethnracehispanic                   3329     4587
ethnraceother                      2877     4226
inr_wave_0_sc                      3491     5155
externalising_wave_0_sc            2399     3651

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

Model 3B longitudinal no interaction effects expressed as % change in expected outcome per 1-unit increase in original predictor scale

```
# A tibble: 2 x 4
  Variable                    Estimate `Lower 95% CI` `Upper 95% CI`
  <chr>                          <dbl>          <dbl>          <dbl>
1 Breast development (Wave 2)     7.19           4.16           10.3
2 ADHD Y                         75.7           61.4            91.5
```

Model 4 longitudinal no interaction output

```
   Estimate Est.Error  Q2.5 Q97.5
R2    0.828    0.0045 0.819 0.836

 Family: poisson
  Links: mu = log
Formula: internalising_wave_3 ~ menarche_status_p_wave_2 + adhd_traits_wave_2_sc + age_wave_2
   Data: analytic_df_wide (Number of observations: 4496)
  Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
         total post-warmup draws = 4000
```

```
Multilevel Hyperparameters:
~family_id (Number of levels: 3915)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.819     0.014    0.792    0.848 1.001     1074     1891

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.109     0.032    0.050    0.177 1.002      520      659

Regression Coefficients:
                          Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS
Intercept                    1.238     0.036    1.164    1.305 1.001     1526
menarche_status_p_wave_2Y    0.172     0.031    0.110    0.231 1.000     1502
adhd_traits_wave_2_sc        0.184     0.013    0.159    0.211 1.000     1089
age_wave_2_sc               -0.011     0.015   -0.041    0.019 1.002     1444
ethnraceblack_nh            -0.416     0.055   -0.522   -0.309 1.000     1423
ethnracehispanic            -0.091     0.048   -0.183    0.005 1.001     1069
ethnraceother               -0.009     0.050   -0.112    0.089 1.001      997
inr_wave_0_sc                0.045     0.018    0.012    0.081 1.001     1220
internalising_wave_0_sc      0.390     0.014    0.362    0.416 1.001      915
                          Tail_ESS
Intercept                     2132
menarche_status_p_wave_2Y     2116
adhd_traits_wave_2_sc         2190
age_wave_2_sc                 2266
ethnraceblack_nh              2381
ethnracehispanic              1677
ethnraceother                 1993
inr_wave_0_sc                 2018
internalising_wave_0_sc       1817

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

Model 4 longitudinal no interaction effects expressed as % change in expected outcome per 1-unit increase in original predictor scale

```
# A tibble: 2 x 4
  Variable            Estimate `Lower 95% CI` `Upper 95% CI`
  <chr>                  <dbl>          <dbl>          <dbl>
1 Menarche (Wave 2)      18.8           11.6           26
```

```
2 ADHD traits (Wave 2)     3.43           2.94           3.93
```

Model 4B longitudinal no interaction output

```
   Estimate Est.Error  Q2.5 Q97.5
R2    0.829   0.00462 0.819 0.837


 Family: poisson
  Links: mu = log
Formula: internalising_wave_3 ~ breast_development_p_wave_2_sc + adhd_traits_wave_2_sc + age_
   Data: analytic_df_wide (Number of observations: 4515)
  Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~family_id (Number of levels: 3928)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.824     0.015    0.796    0.853 1.004      909     1553

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.110     0.032    0.054    0.179 1.002      489      848

Regression Coefficients:
                               Estimate Est.Error l-95% CI u-95% CI  Rhat
Intercept                         1.307     0.034    1.238    1.372 1.000
breast_development_p_wave_2_sc    0.079     0.015    0.050    0.109 1.002
adhd_traits_wave_2_sc             0.182     0.013    0.156    0.207 1.002
age_wave_2_sc                     0.003     0.015   -0.026    0.031 1.000
ethnraceblack_nh                 -0.405     0.054   -0.515   -0.303 1.000
ethnracehispanic                 -0.074     0.047   -0.165    0.020 1.000
ethnraceother                    -0.023     0.051   -0.124    0.078 1.002
inr_wave_0_sc                     0.036     0.016    0.004    0.069 1.001
internalising_wave_0_sc           0.389     0.014    0.361    0.415 1.003
                               Bulk_ESS Tail_ESS
Intercept                           963     1665
breast_development_p_wave_2_sc     1044     1934
adhd_traits_wave_2_sc               859     1433
age_wave_2_sc                      1142     2235
ethnraceblack_nh                    921     1758
ethnracehispanic                    733     1529
ethnraceother                       706     1398
```

```
inr_wave_0_sc                      706     1288
internalising_wave_0_sc            715     1122

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

Model 4B longitudinal no interaction effects expressed as % change in expected outcome per 1-unit increase in original predictor scale

```
# A tibble: 2 x 4
  Variable                    Estimate `Lower 95% CI` `Upper 95% CI`
  <chr>                          <dbl>          <dbl>          <dbl>
1 Breast development (Wave 2)     6.3            3.91           8.76
2 ADHD traits (Wave 2)            3.39           2.9            3.86
```

Model 5 longitudinal no interaction output

```
   Estimate Est.Error  Q2.5 Q97.5
R2    0.819   0.00538 0.808 0.829


 Family: poisson
  Links: mu = log
Formula: externalising_wave_3 ~ menarche_status_p_wave_2 + adhd_traits_wave_2_sc + age_wave_
   Data: analytic_df_wide (Number of observations: 4496)
  Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
         total post-warmup draws = 8000

Multilevel Hyperparameters:
~family_id (Number of levels: 3915)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.976     0.020    0.937    1.016 1.000     2063     4103

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.069     0.037    0.006    0.146 1.001      644     1574

Regression Coefficients:
                          Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS
Intercept                    0.462     0.037    0.387    0.532 1.001     2834
menarche_status_p_wave_2Y    0.120     0.040    0.042    0.201 1.002     2988
```

```
adhd_traits_wave_2_sc        0.292     0.017    0.259    0.326 1.000     2631
age_wave_2_sc               -0.052     0.019   -0.089   -0.014 1.001     2765
ethnraceblack_nh            -0.114     0.065   -0.243    0.013 1.001     2322
ethnracehispanic            -0.101     0.056   -0.211    0.007 1.000     2332
ethnraceother               -0.058     0.064   -0.181    0.070 1.000     2407
inr_wave_0_sc                0.005     0.021   -0.037    0.047 1.000     2810
externalising_wave_0_sc      0.458     0.017    0.425    0.492 1.000     2091
                          Tail_ESS
Intercept                     4499
menarche_status_p_wave_2Y     4440
adhd_traits_wave_2_sc         3956
age_wave_2_sc                 5226
ethnraceblack_nh              3599
ethnracehispanic              4332
ethnraceother                 4024
inr_wave_0_sc                 4206
externalising_wave_0_sc       3695

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

Model 5 longitudinal no interaction effects expressed as % change in expected outcome per 1-unit increase in original predictor scale

```
# A tibble: 2 x 4
  Variable             Estimate `Lower 95% CI` `Upper 95% CI`
  <chr>                   <dbl>          <dbl>          <dbl>
1 Menarche (Wave 2)      12.8            4.29          22.3
2 ADHD traits (Wave 2)    5.49           4.84           6.14
```